

How version our model and datasets?

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MLSECOPS IN ACTION COURSE

How in MLflow we keep track of our training Data?

```
# 🚩 Log the training dataset
mlflow.log_param("dataset_name", "wine_training_data")
mlflow.log_param("dataset_source", "data/train.csv")
mlflow.log_param("dataset_version", "v1.0")
mlflow.log_param("dataset_size", len(train))
mlflow.log_param("dataset_shape", f"{train.shape[0]}x{train.shape[1]}")
```

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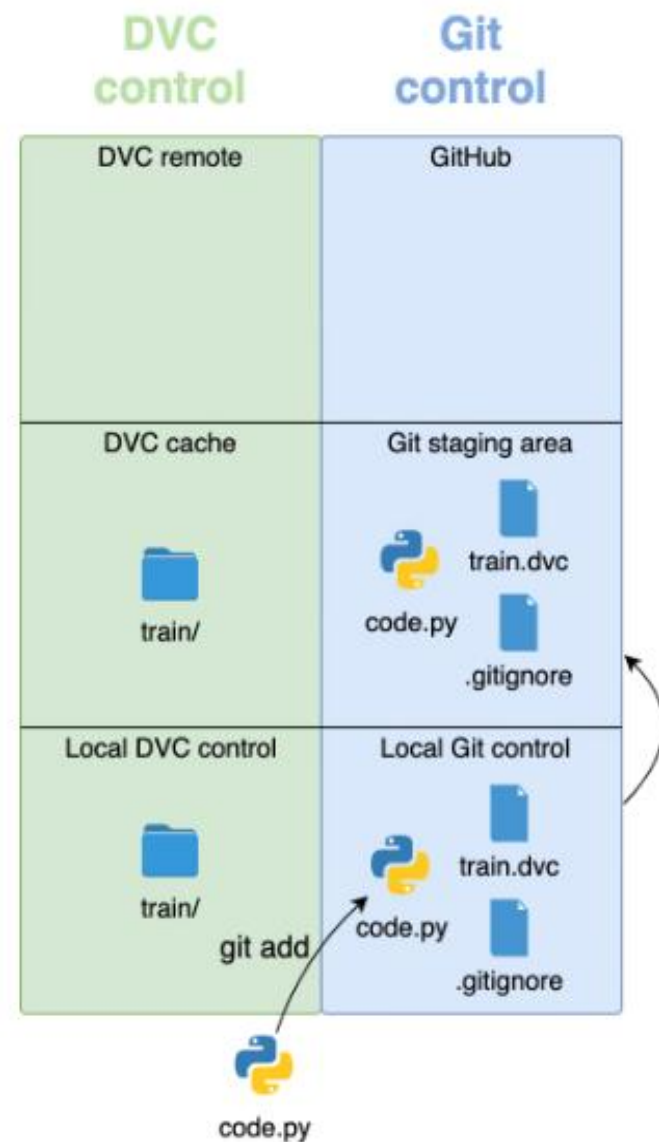
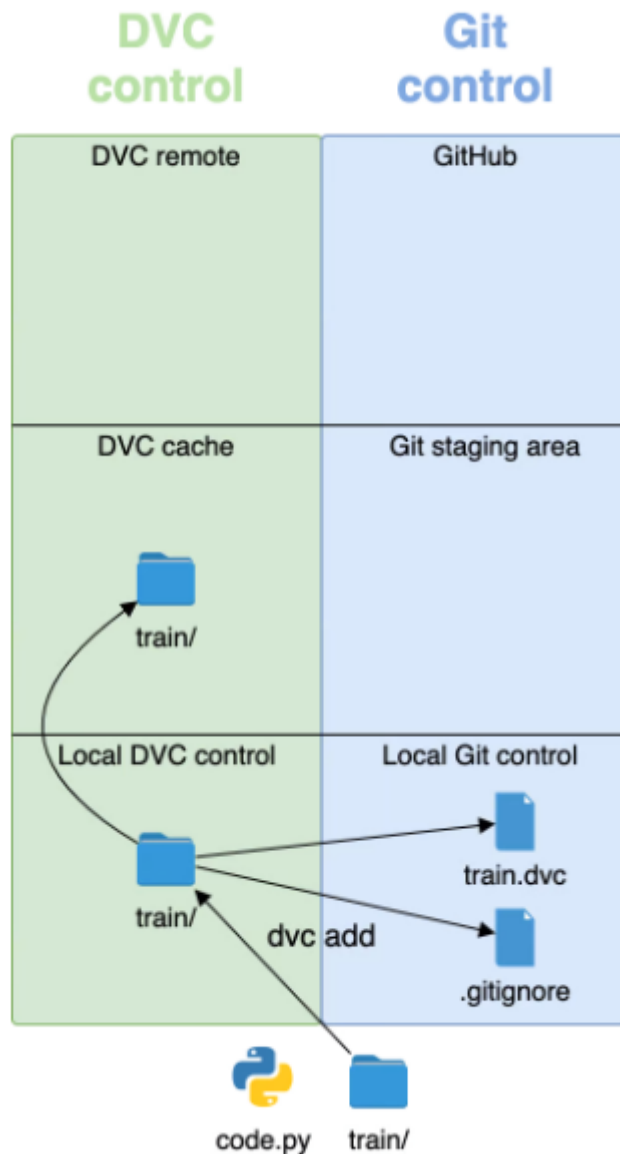
But it is not always GOOD enough and make us TIRED.

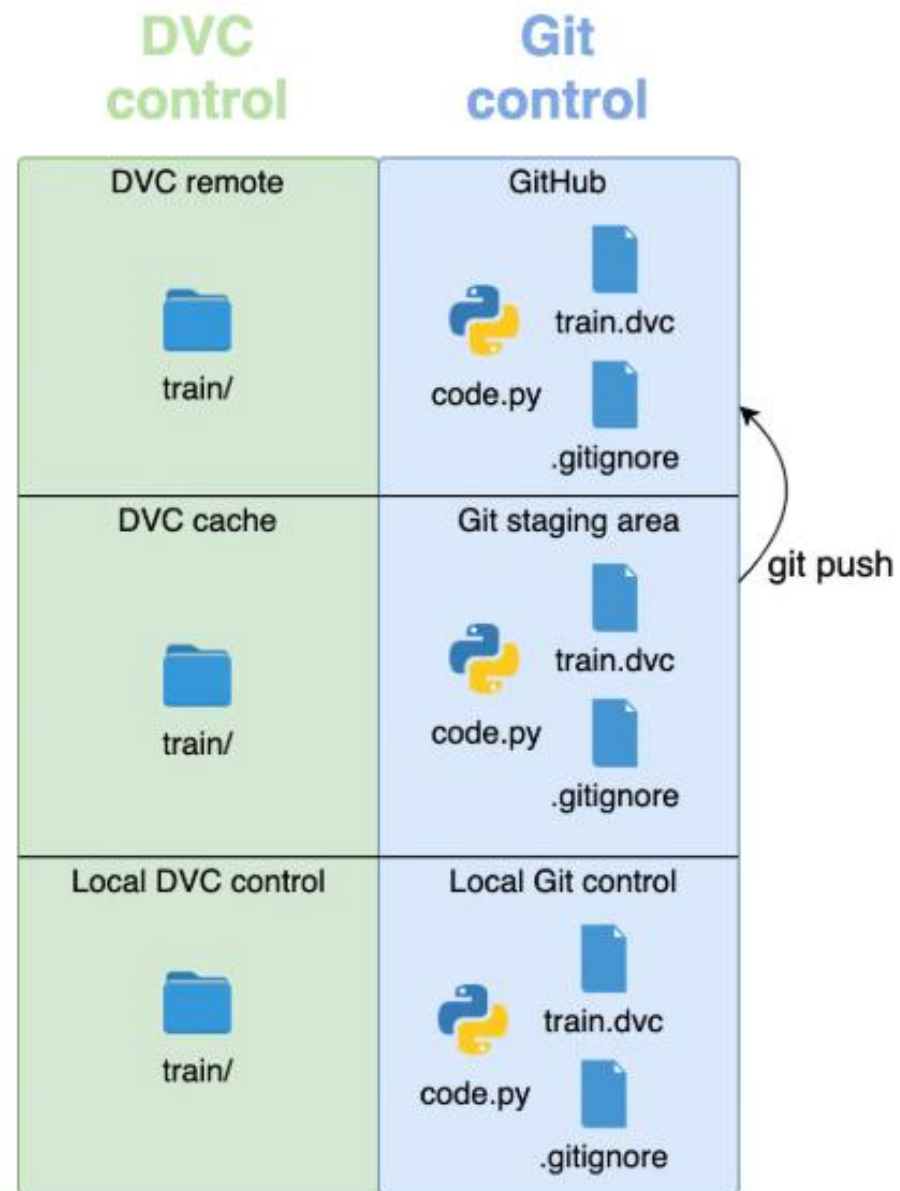
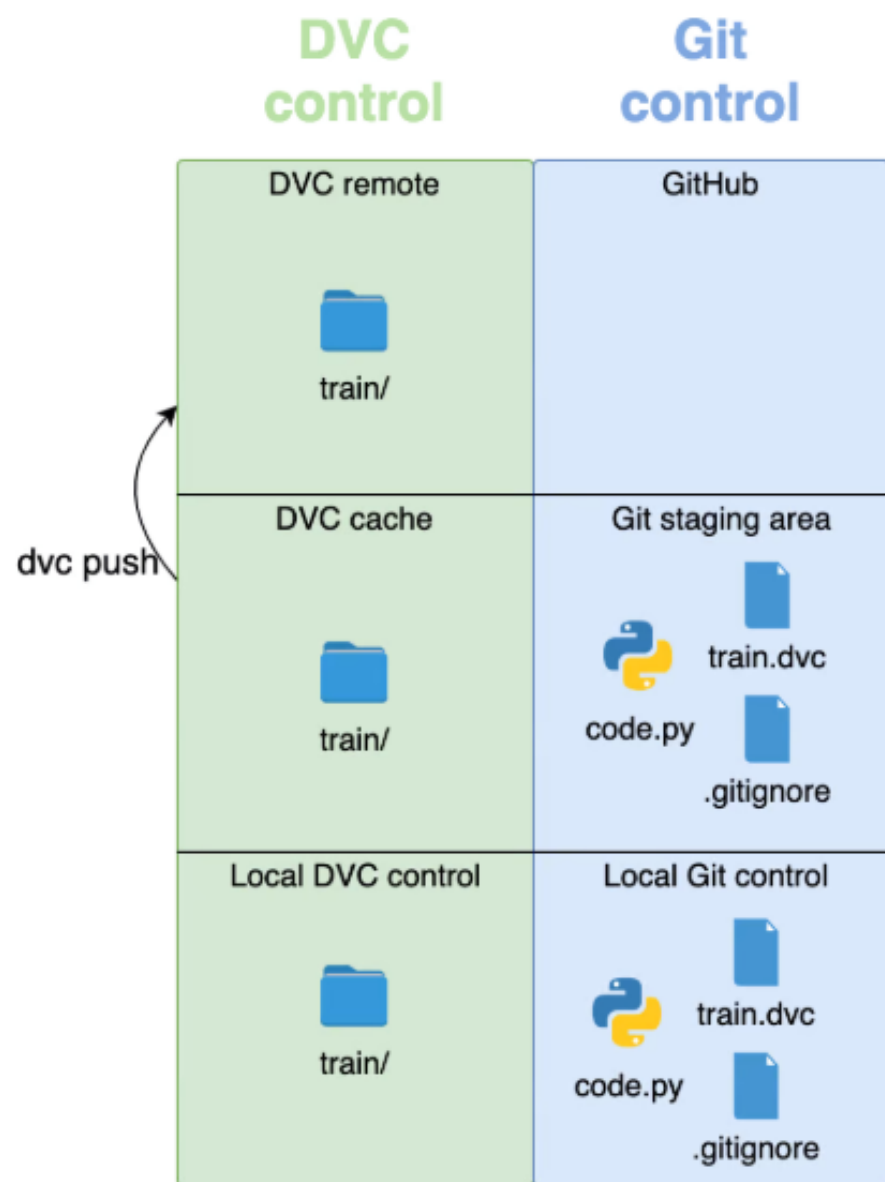
We should automate the Data versioning process.

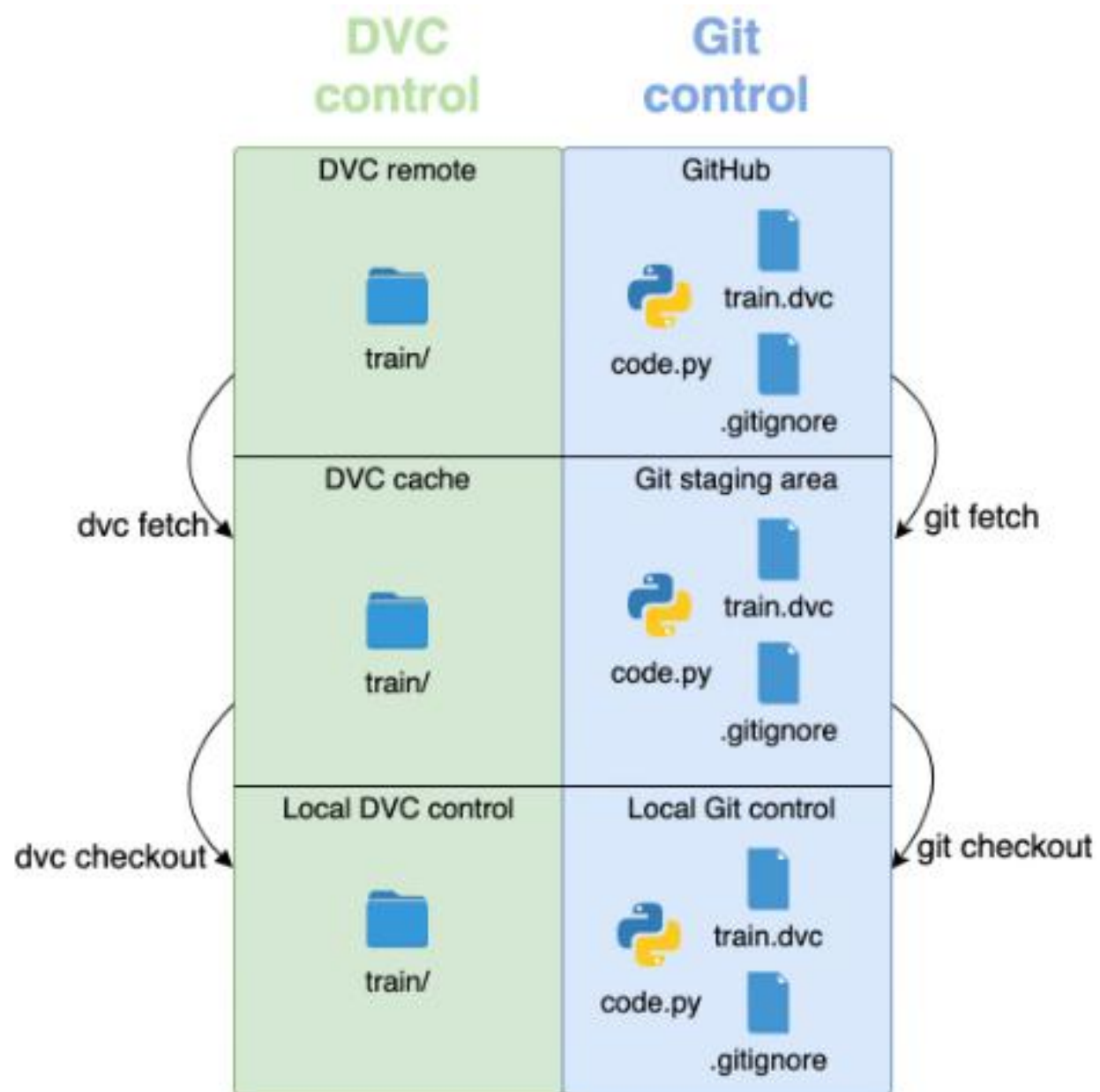


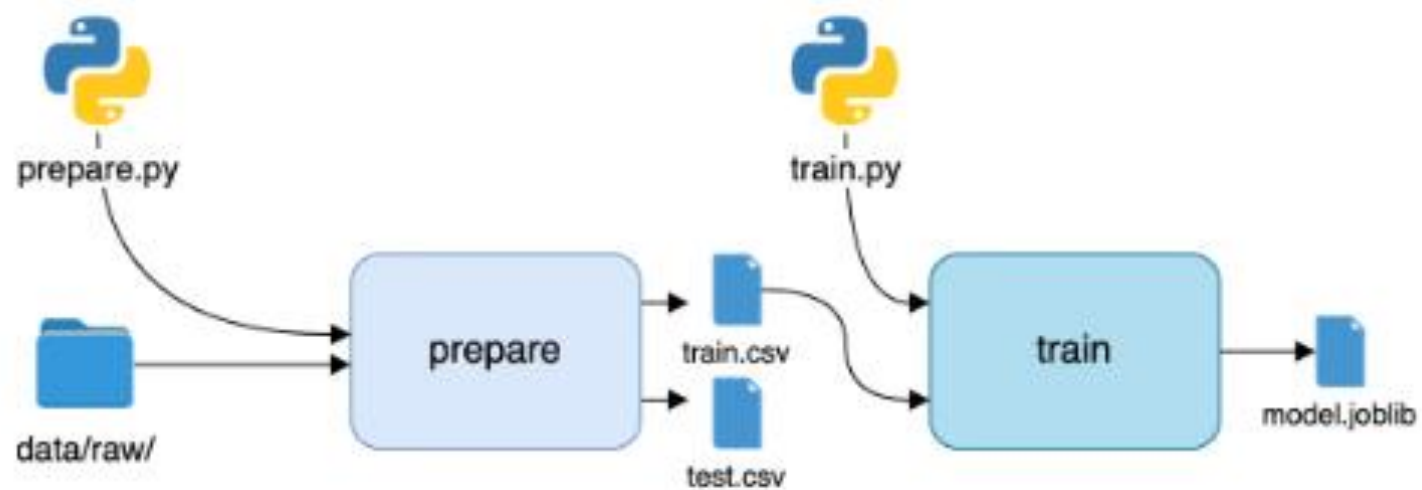
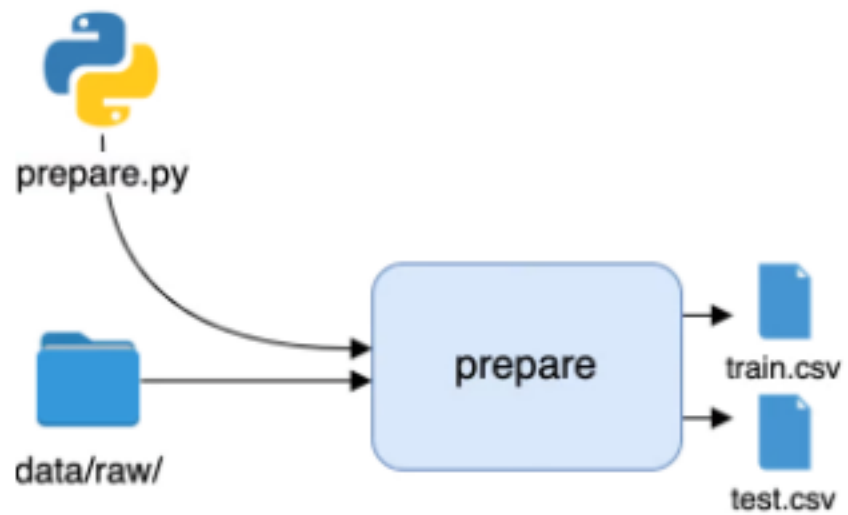
Pachyderm
version-controlled data science

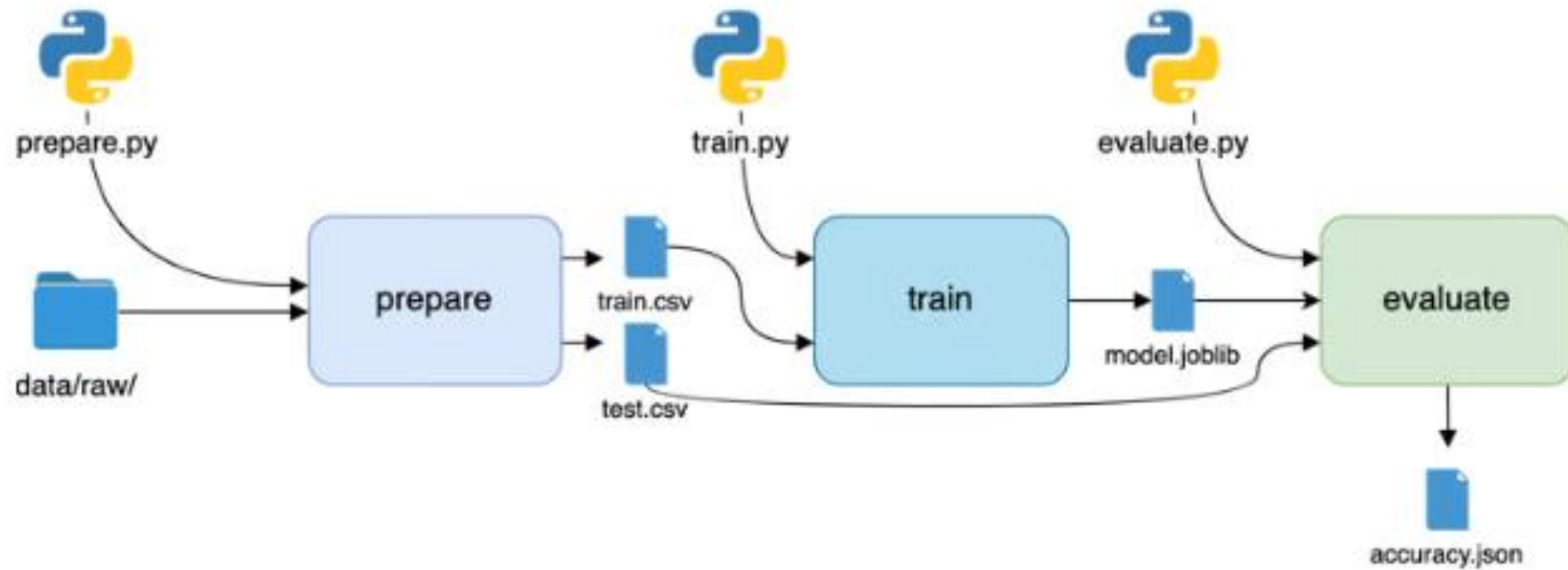




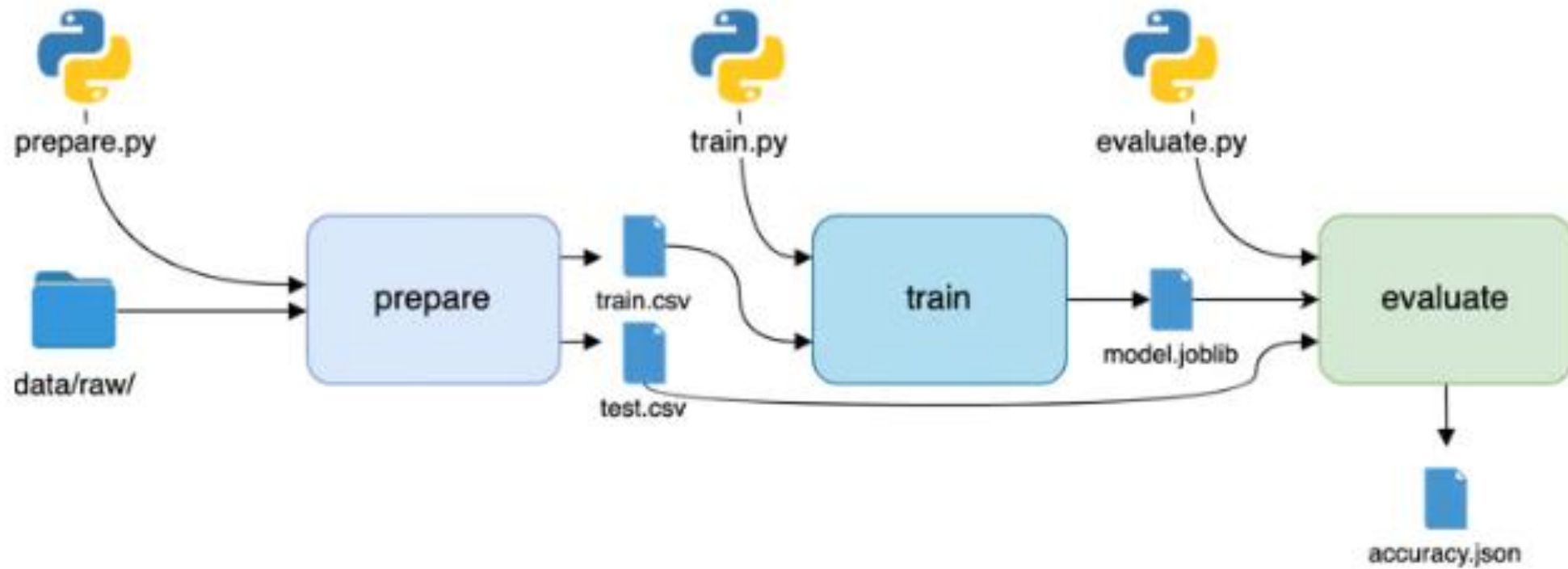








<https://realpython.com/python-data-version-control/>



We will show you how we can automate these steps using DVC

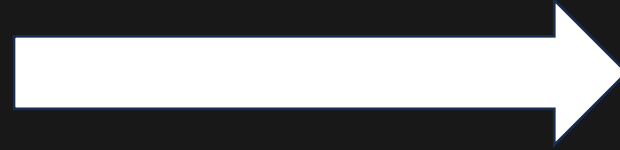
MLflow vs DVC

Feature	MLflow	DVC
Data Versioning	Manual logging	✓ Automatic
Large File Handling	Limited	✓ Efficient
Pipeline Dependencies	Manual	✓ Automatic
Data Rollback	Manual	✓ Automatic with Git
Storage Optimization	Duplicates files	✓ Deduplication

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So, the best approach is **using both DVC and MLflow side by side** for daily ML tasks.



```
+-----+
| create_dataset |
+-----+
***
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+-----+
| train_model |
+-----+
***
***
**
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**
**
+-----+
| eval_model |
+-----+
```



```
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| create_dataset |  
+-----+
```

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```
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```

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```
+-----+  
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+-----+
```


✅ = DVC and 🚩 = MLflow

Data Preparation: Use DVC to prepare and version your data (✅)

Training: Run your training script (DVC repro), then logging metrics and parameters to MLflow (✅, 🚩).

Evaluation: Evaluate your model (DVC repro) and log the results to MLflow (✅, 🚩).

Model Registry: Register the trained model in MLflow for versioning and deployment (🚩).

Reproducibility: If you need to retrain or reproduce an experiment, use DVC to rerun the necessary pipeline stages and log new results to MLflow (✅, 🚩).

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```

Let's go for coding



```
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